

Archit Jambhule

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[Email](#) | [LinkedIn](#) | [Portfolio](#) | [Github](#)

EXPERIENCE

VoicingAI | AI/ML Engineer

Present

- Engineered a **multi-provider voice AI orchestration platform** for **real-time telephony**, enabling full-duplex, low-latency STT-LLM-TTS streaming, distributed state management, and **high-throughput SIP routing**.
- Fine-tuned Whisper and Conformer STT on Indian dialects cutting WER by [X]%; fine-tuned Qwen for tool-calling with [X]% gain on the TAU benchmark.
- Curated **900-hour synthetic multilingual dataset across 31 languages (Indian + European)** powering in-house STT, TTS and Speech to Speech translation model finetuning.
- Deployed voice agent pipelines for **Philips, LTM, LT Realty, SURA Insurance, Anywhere RE with full prod ownership**; orchestration engine achieving **sub-800ms time-to-first-audio**.
- Shipped **AgentforAgents** — multi-agent system (8 agents) that designs and deploys complete voice agent flows via chat, **reducing agent setup time from hours to under 5 minutes**.

Samespace | SDE AI/ML

Mar 2024 - Mar 2025

- Built and deployed **Sphere**, a scalable RAG-based document intelligence platform with integrations to vector and graph databases like **Qdrant**, and **Neo4j**, achieving 95% retrieval precision.
- Selected and fine-tuned open-source models using **QLoRA** and **PEFT with instruction tuning** on domain-specific and synthetic datasets, improving **task-specific accuracy by 65%**.
- Developed an **multi-agent AI system** using **LangChain** and **SWARM**, enabling dynamic handling of user queries across voice calls, text messages and more, achieving a query resolution success rate of 92%.

PROJECTS

Hermes PM

April 2026

- Built **Python package manager with uv-inspired optimizations** (PEP 658 metadata, PubGrub solver, reflink installs, parallel asyncio) **achieving 1.78x faster installs and 30% disk savings vs pip**
- Engineered **parallel download/install pipeline** delivering **4-12x speedup from cold to warm cache on APFS/Btrfs filesystems**
- Implemented **automated security auditing via OSV vulnerability database** with orphaned dependency cleanup, and deterministic lock files for reproducible installs

Real-Time Object Detection with PaliGemma | Python, PyTorch, OpenCV

January 2023

- Built an **object detection system using the PaliGemma 3B model** achieving real-time processing for videos.
- Processed high-resolution videos with adaptive bounding boxes and **95% accuracy in detecting user-specified objects**.
- Supported **diverse media inputs** (URLs, images, videos) and **automated output generation**.

SKILLS

Languages: Python, Golang, TypeScript, SQL

AI/ML: LLMs, RAG, Multi-Agent Systems, Finetuning (QLoRA, PEFT), STT, TTS, Speech-to-Speech, Hugging Face, PyTorch, Transformers, Langgraph

Databases: QdrantDB, Postgres, Redis, Pinecone

Infrastructure: MCP, gRPC, WebSockets, SIP, Docker, Kubernetes, AWS, GCP, CI/CD, MLOps, Prometheus, Grafana, Git

EDUCATION

Indian Institute of Technology, Roorkee (IIT Roorkee)

Uttarakhand, India

Master's in Biotechnology (Computational Biology), CGPA : 8.72

2021 – 2023